

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Built and shipped an autonomous Jira ticket-resolution agent that automated operations for business stakeholders: 6-gate triage, MCP-driven context gathering, agentic RAG with FAISS + BM25 retrieval over 11,000 historical SQL queries, and self-healing execution with a 10-retry budget. Closed 400+ data requests across 8 categories over 15 months and cut routine turnaround from 30–60 minutes to under 10 minutes each. Stakeholders never knew an agent was answering them.
- Designed and shipped to production a self-service analytics agent over 67 BigQuery tables with sub-agent orchestration and human-in-the-loop validation (Context Researcher → SQL Drafter → Validator → Devil's Advocate); 8 deterministic SQL rules and a 23-case golden-query evaluation suite. Approved by Sr. Director and product leadership and currently the only AI skill in active use by business teams on Data Ventures. Partnered directly with Product, Data Science, and business stakeholders to turn ambiguous asks into agent-ready specs.
- Built and deployed a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; end-to-end observability via decision logging and BigQuery state persistence, so the agent has no LLM-memory reliance. Designed a 3-tier event-driven v2 platform self-configuring via a BigQuery event registry — new event types ship without code deploys.
- Built a registry-governed semantic layer over BigQuery that boxes the LLM into semantic parsing only — one LLM call maps the question to strict JSON intent (the model never writes SQL), a deterministic gate validates and canonicalizes intent, and a deterministic compiler emits the SQL — so the same question returns the same answer every time and the one probabilistic step stays regression-testable. Established the guardrails that keep AI-generated queries correct and production-ready.
- Replaced ~20 distinct recruitment workflows I'd been running by hand 1–2× weekly with end-to-end automation on a 68-table BigQuery platform. Stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates. A recurring 3-hour task now takes ~3 minutes of setup.
- Delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users); cut data-request turnaround from 1–24 hours to 5–10 minutes; Making a Difference Award.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- RAG-to-proposal generator deployed on AWS EC2; cut proposal time ~30%; Best in Show — UChicago MSADS Capstone.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit + Gemini 2.5 + LangChain-orchestrated RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Agent stack: MCP (Model Context Protocol) server design, sub-agent orchestration, context engineering, prompt engineering, RAG (FAISS + BM25 + RRF), evaluation suites (golden sets, human-in-the-loop validation), multi-model routing across providers, headless cron agents, event-driven architectures, decision logging / observability

Cloud / data: AWS (EC2), BigQuery, Databricks, dbt, SQL, Python, PySpark, Airflow, MySQL

LLMs: Claude (Opus / Sonnet / Haiku via Wibey), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

APIs / triggers / glue: FastAPI, Cloud Run, BigQuery scheduled queries, Slack webhooks, cron, Playwright SSO, JSONL state stores, CI/CD (pytest, version control)

Viz: Power BI (DAX, XSLT), Tableau, LookML, Streamlit, Matplotlib, React Recharts

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL